

PRAJWAL M

COMPUTER SCIENCE STUDENT

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CAREER SUMMARY

Computer Science student focused on software development, backend systems, network security, and system automation. Experienced in Java and Python with hands-on expertise in designing multi-threaded, low-latency systems. Proven ability to build reliable and efficient cross-platform solutions, supported by published research in system-level automation.

EDUCATION

- **Bachelor of Engineering (B.E.), Computer Science and Engineering**
Sri Sairam College of Engineering, VTU 2023-26
- **Diploma in Mechtronics Engineering**
Sri Jayachamarajendra Polytechnic, Department of Technical Education 2020-23

TECHNICAL SKILLS

- **Programming Languages:** Java, Python, C, C#, SQL
- **Frameworks & Tools:** Flask, Git, VS Code, Postman
- **Databases:** MySQL, MongoDB, SQLite
- **Web Technologies:** HTML, CSS, JavaScript

PROJECTS

Automated Student Attendance and behavioral System Repository

- Designed and deployed a **Flask**-based attendance automation application leveraging image-based verification and **deep-learning**-powered face recognition.
- Engineered a scalable backend using **Python**, **OpenCV**, **MTCNN**, and **InsightFace** for data processing, feature extraction, and matching against a pre-stored student database.
- Utilized **NumPy** and **Pandas** for data handling, preprocessing, and computational workflows.
- **Technologies Used:** Python, Flask, OpenCV, MTCNN, InsightFace, NumPy, SciPy, Pandas, SQLite.

Network-Driven Automated Software Deployment and Installation Repository

- Architected a secure master-slave automation system using Python **Sockets** with both **UDP** and **TCP** protocols and **Multithreading** to manage remote software Installation, reducing manual software deployment time by 80% across networked environments.
- Implemented **End-to-End Encryption** using **TLS** wrappers and **PBKDF2** hashing for secure authentication and command transmission. File integrity is verified using the **SHA-256** algorithm. **Salted passwords** for saving Passwords.
- Developed a **Tkinter-based GUI** for the centralized Master node and utilized **SQLite** to maintain real-time inventory and client status logs.
- Engineered a persistent **Windows Service** using **pywin32** and **subprocess** to execute silent installations (.exe, .msi) and remote power management commands with SYSTEM privileges.
- **Technologies Used:** Python, Tkinter, Socket (TCP/UDP), TLS, SQLite, pywin32, psutil, pyinstaller.

Secure Unified Desktop Administration System (SUDAS) [Repository](#)

- Engineered a real-time client–server Architecture connecting **Android (Java)** and **Windows (Python)** for low-latency remote desktop interaction, supporting **mouse** control, **keyboard** input, **presenter**-mode tools using UDP sockets.
- Designed and implemented **gesture**-based control on the mobile client, mapping touch and motion gestures to system-level mouse events on the host machine.
- Implemented secure communication using **AES-256** encryption and **HMAC-SHA256** to ensure data confidentiality and integrity, with **timestamp**-based replay attack mitigation.
- Designed a dual-protocol architecture using **UDP** for latency-sensitive input and streaming **Live Preview**, and **TCP** for reliable **file transfer** and large payload delivery.
- Developed a Python watchdog service to monitor connection stability using **Heartbeat Mechanism**, handle automatic recovery, and enable device discovery via UDP broadcast on local networks.
- **Technologies Used:** Java (Android), Python, Socket (TCP/UDP), AES-256, HMAC Security, PyAutoGUI, Pillow Multithreading.

PUBLICATIONS

- “Smart Network-Based Automated Software Deployment for Multi-System Lab Environments” IJIRIS Journal, Vol. 11, Issue 08, Nov 2025 <https://doi.org/10.26562/ijiris.2025.v1108.25>

CERTIFICATE

- **Simplilearn- Python for Beginners** – Practical, hands-on experience with programming fundamentals, OOP concepts, multithreading, scripting, data visualization, and essential libraries.
- **Simplilearn- Java Programming for Beginners** – Practical, hands-on experience in Java, covering core fundamentals and OOPs concepts.
- **NPTEL- Internet of Things** – Knowledge of Internet of Things concepts, including device connectivity, sensor data processing, and basic automation.
- **Infosys Certification in Principles of Generative AI** – Introduced AI models, deep learning techniques, hands-on Modern AI application, overview on ML.